Decoding Body Transformation Experiences: Fusing AI-Ready Wearables and Haptic Illusions in Everyday Settings

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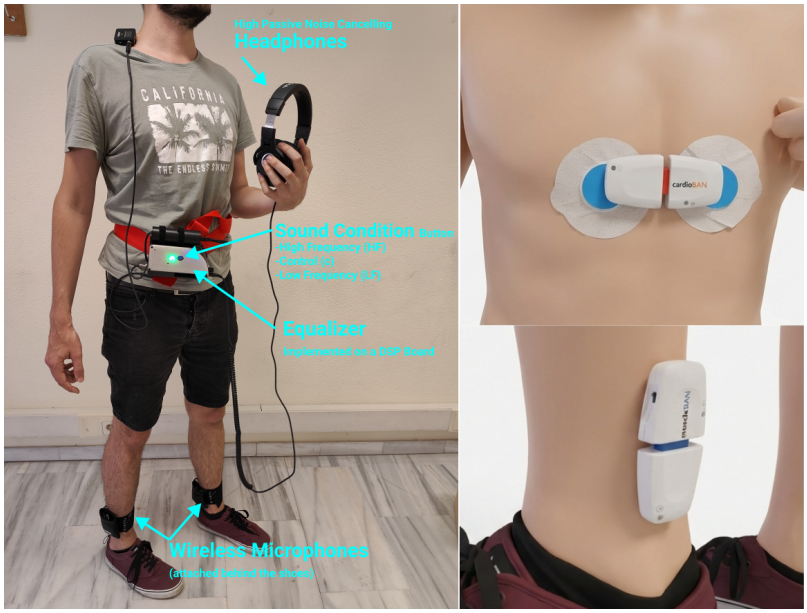

Fig. 1. SoniWeight Shoes System. (Left) User wearing the prototype: wireless microphones on ankles, and the DSP unit (Bela) on the waist delivering audio via headphones. (Right) Sensor placement: Cardioban (ECG) on the chest and Muscleban (EMG) on the calf.

Body perception is fundamentally malleable and can be altered through interactive multisensory feedback. We are a team of researchers exploring the intersection of wearable computing, haptics, and body perception. In this position paper, we introduce our recent work on auditory Body Transformation Experiences in everyday settings and a novel multi-muscle haptic prototype (VibrOut) designed to alter the perceived spatial dimensions of the body. We leverage this work as context to propose a discussion on how longitudinal multimodal data and novel haptic actuators can be used to train AI models not for actuation, but for deep behavioral analysis as to provide a real time proxy of a person's current body perception. This in turn may inform the selection of stimuli to get to specific states. We seek to discuss with the workshop community the engineering and data challenges of creating human-centered AI monitoring systems capable of decoding physiological and kinematic patterns to infer emotional states and perceptual shifts in everyday life.

Additional Key Words and Phrases: Wearables, Body Perception, Body Transformation Experiences, Haptics, Auditory Feedback, Affective Computing, Field Study

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1 Introduction

We are a collaborative research team based at Universidad Carlos III de Madrid, working at the intersection of Human-Computer Interaction, multisensory feedback, and wearable computing. Our core research focuses on engineering and validating Body Transformation Experiences (BTEs)—interventions that use sensory feedback to alter how people perceive their own bodies (e.g., feelings of weight, strength, or spatial volume) [7, 10].

Historically, highly controlled laboratory settings have been required to reliably induce these illusions, as they demand ultra-low latency to preserve agency and sensorimotor integration [2, 12, 17]. However, our recent efforts focus on transitioning these technologies into unsupervised, everyday environments [13–15]. By participating in the *AI for Haptics and Haptics for AI* workshop, we hope to connect with experts in machine learning and haptic design. We believe that integrating our in-the-field multimodal physiological tracking with novel haptic actuators can provide the foundation for AI models that intelligently analyze the effects of BTEs.

To provide context for our discussion, we briefly present two complementary research threads. First, we recently engineered a standalone wearable architecture (shown in Figure 1) capable of delivering ultra-low latency (1.7 ms) auditory feedback to alter perceived body weight and strength (the *Footsteps Illusion*) [4, 18, 19]. Built upon the Bela platform [11], we deployed this system in a four-week longitudinal home study ($N = 18$), validating that these illusions persist in the field without habituation [5]. More importantly for the AI community, this system simultaneously captures synchronized kinematics, Electrocardiogram (ECG), and Electromyogram (EMG) [1, 3] alongside subjective labels. This yields an "AI-Ready" multimodal dataset, capturing the raw physiological ground-truth of a user experiencing altered body perception.

2 Problem Framing and Position Statement

Parallel to auditory interventions, we are exploring haptics to alter the perceived spatial dimensions of the body. Previous research has shown that vibrating a single muscle tendon can induce illusory movement or perceived changes in limb dimensions [6, 8]. Building on this, we developed *VibrOut* (illustrated in Figure 2), a novel prototype involving simultaneous vibration of multiple synergist muscles across both sides of the body [21].

By targeting muscles involved in arm abduction while the arms remain stationary, VibrOut activates primary afferents in muscle spindles [20], creating a powerful illusion that the arms are lifting laterally. This alters the perceived size of elements in the surrounding space and bodily interactions with the environment.

By combining the AI-ready physiological tracking of our longitudinal wearable with the perceptual modulation of our VibrOut haptic prototype, we envision a comprehensive AI monitoring framework for decoding Haptic-Driven Perceptual States. Rather than using AI to act on the haptics or directly modulate perception—which is handled securely by our prototypes—our position is to leverage AI to analyze the multisensory effects and find patterns in users' movements. For example, by analyzing kinematics and physiological responses, an AI system could reliably detect if a person is feeling sad, lethargic, or experiencing a shrinking body image based on their movement signatures [9, 16]. This in turn could inform the selection of haptic stimuli to induce specific perceptual states.

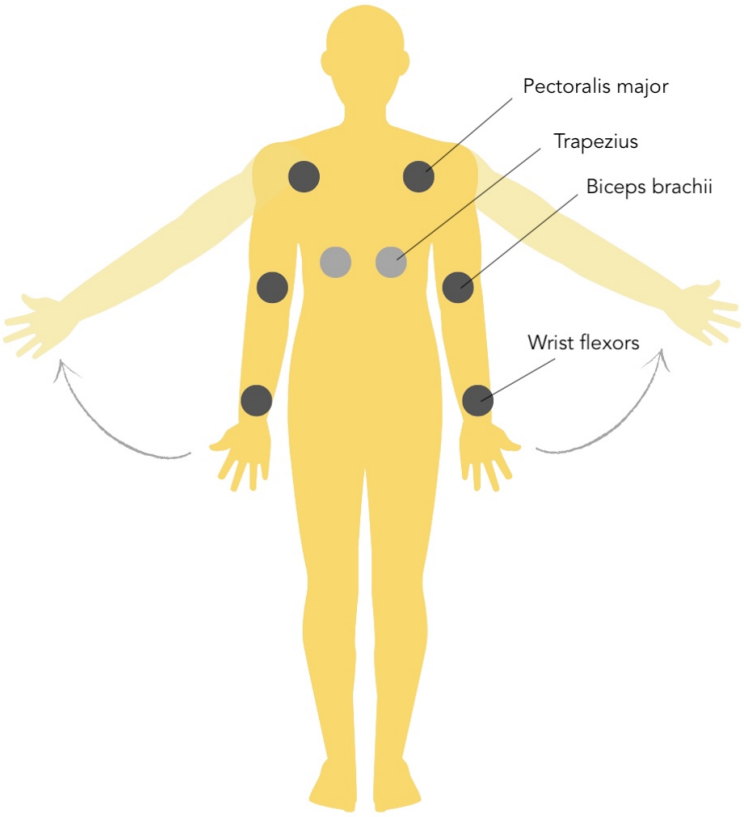

Fig. 2. VibrOut Prototype. The hardware consists of 8 speaker driver exciters connected to an amplifier and computer. Max/MSP software triggers a simultaneous 70Hz (6dB) vibration across multiple synergist muscles, creating the illusion of lateral arm abduction and expanded spatial volume.

3 Implications and Future Directions

To realize this vision, we believe the following topics require deep discussion within the workshop community:

- **Multimodal Sensor Fusion for Perceptual and Affective Monitoring:** How can we best leverage longitudinal, noisy physiological datasets (ECG, EMG, Kinematics) gathered in the field to train AI models? Specifically, what architectures are best suited to map subtle kinematic changes and biosignals to complex emotional states (e.g., sadness) and implicit perceptual shifts?
- **Haptic Deployment & Analytical Challenges:** What are the hardware and compliance constraints of transitioning multi-muscle haptic actuators (like VibrOut) from the lab to unsupervised domestic environments? Furthermore, how can AI analytical tools help us better quantify and understand phenomena like sensory habituation over longitudinal field use?

4 Conclusion

We look forward to presenting these prototypes and discussing these open challenges with the community. By combining robust haptic and auditory interventions with advanced, multimodal AI analysis, we can move closer to creating human-centered monitoring systems capable of decoding and supporting human perception and emotion in everyday life.

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The Authors

The authors form a collaborative research team based at Universidad Carlos III de Madrid (UC3M) in Spain, specializing in Human-Computer Interaction, multisensory feedback, and wearable computing. Led by Ana Tajadura-Jiménez, the team includes Tomás Martínez-Cortés, Karunya Srinivasan, and Amar D'Adamo. Their collective expertise spans the engineering and validation of Body Transformation Experiences (BTEs), focusing on transitioning highly controlled laboratory haptic and auditory illusions into unsupervised, everyday environments to understand and modulate body perception.